

Jianlun (Lody) Sun, MSc.

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Research Interest

I aim to study how the brain uses predictive, hierarchical structure to construct meaning from experience, from prediction-error dynamics to individual and clinical variation in how predictive models are built and updated. My work combines behavioral modeling, EEG, and naturalistic paradigms. For my PhD, I hope to bridge the mechanistic and the naturalistic, connecting how predictive models are built at the level of neural dynamics to how they shape meaning and vary across individuals and mental states, through a multimodal approach.

Education

Sep 2024 – Jun 2026

📖 **M.Sc. Neuroscience, University of Helsinki** (GPA: 4.5/5)

Thesis title: *Staged Rule Acquisition in WishGLD: Dissociation and Compression Across Learning*.

- Developed a trial-resolved analysis framework for a naturalistic set-shifting task (WishGLD), combining behavioral modeling, two-state HMM trajectory markers, and feedback-locked EEG to characterize how rule acquisition unfolds from search to stabilized performance.
- Identified that rule acquisition is not a unitary event: logical hypothesis collapse, first correct response related neural confirmation, and later behavioral stabilization can dissociate, with early learning showing a clearer staged transition structure that becomes compressed with task exposure.

Sep 2018 – Jun 2022

📖 **B.Sc. Biological Sciences, ShanghaiTech University**.

Thesis title: *Exploring the relationship between Locus Coeruleus integrity and set-shifting dysfunction in Parkinson's Disease*.

Research Publications

Conference Presentations

- 1 J. Sun, K. Lukander, and B. U. Cowley, "Staged rule acquisition in WishGLD: Dissociation and compression across learning," in *9th International Conference on Motivational and Cognitive Control (MCC)*, Poster presentation (accepted), Helsinki, Finland, Aug. 2026.

Research Experience

Jun 2025 – Jun 2026

📖 **Research Assistant**, HiPerCog Lab, Department of Education, University of Helsinki (Supervisor: Ben Cowley).

- Designed and implemented a modified WCST in PsyToolkit with an orthogonal orientation rule, enabling future reinforcement-learning modeling of multidimensional strategy use.

Research Experience (continued)

- Mar 2021 - Jun 2022
- Undergraduate Research Assistant**, Institute of Neuroscience, Chinese Academy of Sciences (Supervisor: Ye Zheng)
- Quantified Locus Coeruleus/Substantia Nigra integrity from neuromelanin-MRI via manual ROI tracing and intensity normalization
 - Linked LC structural metrics to set-shifting deficits in Parkinson’s disease (CANTAB IED)
 - Assisted in collecting and organizing neuropsychological data from DBS patients, contributing to analyses of stimulation-related cognitive impairment in GPi/STN cohorts. (Collaborator: Dr. Wu Jianjun)

Skills

- Languages
- Mandarin (Native), English [TOEFL iBT: 112/120 (R28, L30, W30, S24)]
- Coding
- Python, R, MATLAB
- Analytical Skills
- Python (pandas, NumPy, scipy, matplotlib);
EEG (MNE-Python, EEGLAB, ERP/time-frequency analysis, ISC, IS-RSA);
Statistics/modeling (GLM, PCA, HMM, mixed-effects models, permutation testing, FDR correction);
fMRI (GLM, MVPA/decoding, RSA - independently implemented)

Academic Awards

- 2024-2026
- 100% Tuition Fee Waiver**, University of Helsinki, Master’s Programme in Neuroscience

Research-related Activities

- Oct 2025 - Jun 2026
- Curriculum Reviewer**, Master’s Programme in Neuroscience, University of Helsinki
- Selected student reviewer for evaluating and redesigning the online course **Systems & Cognitive Neuroscience**.
 - Provided pedagogical feedback on structure, clarity, and accessibility for students with diverse academic backgrounds.
 - Contributed to improving learning alignment and conceptual scaffolding for future cohorts.
- Jan 2025 - Jun 2026
- Student Representative**, Master’s Programme in Neuroscience, University of Helsinki
- Served as student representative on the programme board; communicated student needs and coordinated curriculum improvements.
 - Synthesized student feedback into actionable recommendations that informed course design, assessment methods, and program structure.